

Total Marks – 35
Date – 4th Sept 2025
Quiz-1
Section – 3J

Total time – 35 mins

Student Roll No. _____
Student Name: _____

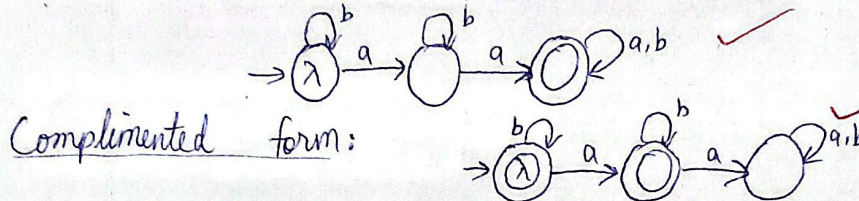
Q:1 - Concatenate as $L_2.L_1$ when $L_1 = \{1, 010, 010\}$ and $L_2 = \{0, 01, 011\}$ (2 marks)

~~$\{01010, 01011\}$~~
 $\{00, 001, 0011, 010, 0101, 01011, 0110, 01101, 011011\}$

Q:2 - How many occurrences of 101 in string 101010101? (1 mark)

4 occurrences

Q:3 - Make an FA state diagram for language accepting at-least 2 a's with $\Sigma = \{a, b\}$ and then compliment it in another diagram. Also state the new (complimented) FA valid and invalid strings (4 each) (2+3+2+2+1 marks)



valid strings: bbbb, ba bb, abbb

invalid strings: aabb, baba, aaaa

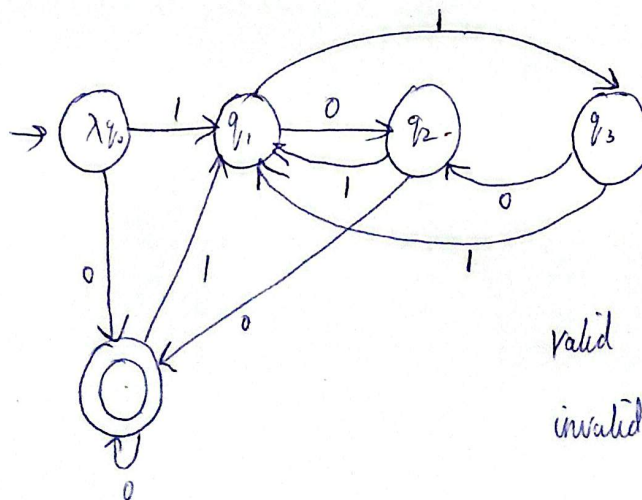
$L = \{x | x \text{ belongs to } \Sigma \text{ and } |a| \text{ in } x \geq 2\}$

$L' = \{x | x \text{ belongs to } \Sigma \text{ and } |a| \text{ in } x \leq 2\}$

Note: write in terms of $<, \leq, >, \geq, =$

$(\because \geq \Rightarrow \leq)$

Q:4 - Make an FA that accepts binary strings that are divisible by 4 if $\Sigma = \{0, 1\}$. Enumerate the language as well. (5+2 marks)



valid strings: $\{0, 100, 1000\}$

invalid strings: $\{10, 011, 101\}$

Q:5 - Let M_1 and M_2 be the FAs shown below, accepting languages L_1 and L_2 , respectively.
 Draw State Diagram for both FAs separately where $\Sigma = \{a, b\}$
 L_1 = All strings with at most two b's
 L_2 = All strings with odd a's

Draw an FA accepting the following language. (3+2+10 marks)

a) $L_1 \cap L_2$

